

## A wavelet-based time–frequency analysis approach for classification of motor imagery for brain–computer interface applications

To cite this article: Lei Qin and Bin He 2005 *J. Neural Eng.* **2** 65

View the [article online](#) for updates and enhancements.

### You may also like

- [Classification of Motor Imagery EEG signals using high resolution time-frequency representations and convolutional neural network](#)  
V Srimadumathi and M Ramasubba Reddy
- [Feature extraction for EEG-based brain–computer interfaces by wavelet packet best basis decomposition](#)  
Bang-hua Yang, Guo-zheng Yan, Rong-guo Yan et al.
- [Direction decoding of imagined hand movements using subject-specific features from parietal EEG](#)  
Gangadharan K Sagila and A P Vinod

**physicsworld** LIVE WEBINAR

sponsored by



## End-to-end dosimetry audit for SRS/SBRT: optional or mandatory at the beginning of the journey?

Experience of RTsafe succeSRS/SBRT implementation for Varian Halcyon machine.

The webinar will present the overall experience of a radiotherapy department that utilized RTsafe's QA solutions in preparation for achieving ISRS certification. The session will focus on the use of RTsafe's Prime phantom in combination with film remote dosimetry services, demonstrating how this approach enables End-to-End QA testing and supports accurate, reproducible film dosimetry audits. Attendees will gain insights into how these tools can be employed to validate the entire SRS treatment workflow - from imaging and planning to dose delivery - while aligning with the rigorous standards required for ISRS certification.

Wed, Jun 17, 2026 2:00 PM - 3:00 PM BST

 [click to register](#)



# A wavelet-based time–frequency analysis approach for classification of motor imagery for brain–computer interface applications

Lei Qin and Bin He

Department of Biomedical Engineering, University of Minnesota, 7-105 BSBE,  
312 Church Street, Minneapolis, MN 55455, USA

E-mail: [binhe@umn.edu](mailto:binhe@umn.edu)

Received 12 April 2005

Accepted for publication 10 June 2005

Published 15 August 2005

Online at [stacks.iop.org/JNE/2/65](http://stacks.iop.org/JNE/2/65)

## Abstract

Electroencephalogram (EEG) recordings during motor imagery tasks are often used as input signals for brain–computer interfaces (BCIs). The translation of these EEG signals to control signals of a device is based on a good classification of various kinds of imagination. We have developed a wavelet-based time–frequency analysis approach for classifying motor imagery tasks. Time–frequency distributions (TFDs) were constructed based on wavelet decomposition and event-related (de)synchronization patterns were extracted from symmetric electrode pairs. The weighted energy difference of the electrode pairs was then compared to classify the imaginary movement. The present method has been tested in nine human subjects and reached an averaged classification rate of 78%. The simplicity of the present technique suggests that it may provide an alternative method for EEG-based BCI applications.

(Some figures in this article are in colour only in the electronic version)

## 1. Introduction

A brain–computer interface (BCI) allows a person to communicate with his environment without using the brain's normal output pathways of peripheral nerves and muscles, thus offering patients who are 'locked-in' (e.g., patients with amyotrophic lateral sclerosis) a new channel to interact with the external world (for a review, see Wolpaw *et al* (2002) or Vallabhaneni *et al* (2005)). In order to achieve this goal, BCIs should be designed to translate brain signals into control commands that would allow subjects to operate various tools, such as a computer, a wheelchair or even a robotic device. Various electroencephalogram (EEG) signals have been used as input to a BCI, such as VEP (Sutter 1984), P300 (Farwell and Donchin 1988), slow cortical potential (Birbaumer and Kubler 2000) and oscillatory EEG components (Wolpaw *et al* 1991, 1997, Pfurtscheller and Neuper 1994, Pfurtscheller *et al* 1997, Babiloni *et al* 2000, 2001, Millán *et al* 2002, Wang and He 2004, Vallabhaneni and He 2004, Wang *et al* 2004).

Two types of oscillations are of importance for BCIs: the mu rhythm (8–12 Hz) and beta rhythm (18–25 Hz) (Pfurtscheller *et al* 1997, Wolpaw *et al* 2002), both originating in the sensorimotor cortex. Motor imagery can cause an event-related desynchronization (ERD) which performs an amplitude suppression or event-related synchronization (ERS) which performs an amplitude enhancement in these two frequency bands. During motor imagery, the mu and beta ERD occur mostly on the contralateral hemisphere at the onset of imagination while the beta ERS occurs on the contralateral hemisphere at the offset of the imagination. This beta ERS is considered as a recovery from desynchronization after termination of the imagination (Pfurtscheller 1999). Thus during motor imagery, the most typical topography is a contralateral mu and beta ERD followed by a beta ERS. These different patterns generated by different hand-movement imagination are used for classification to give simple binary control signals for BCI. Recently, there are several

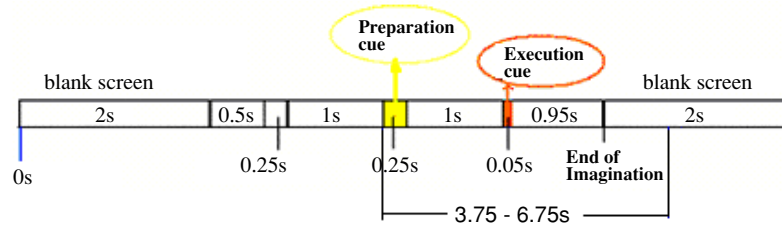


Figure 1. Time sequence of one trial epoch of the experiment.

methods being reported for this kind of classification. An autoregressive (AR) model was applied to short overlapping EEG segments. From the AR spectrum, band power is calculated in several frequency bands and the power sum is used as independent variables in a linear function that defines the control signals (McFarland *et al* 1997a, 1997b, 2000). In a different approach, the most reactive frequency bands for each subject were selected by using the distinction sensitive learning vector quantization (DSLQ) (Kohonen 1990). For the selected bands, the band power is calculated and classified by a nonlinear classifier (LVQ) (Kalcher *et al* 1996, Pfurtscheller *et al* 1997). Common spatial patterns (CSP) combined with simple linear or quadratic Bayes classifiers were another method (Muller-Gerking *et al* 1999). Recently, algorithms using weighted time–frequency analysis (Wang and He 2004, Wang *et al* 2004) were developed.

Time–frequency (TF) analysis is used to describe the distribution of signal energy as a function of both time and frequency. It provides a powerful tool for nonstationary signal processing and has been used in EEG signal analysis. Sinkkonen *et al* showed that the TF analysis provides an informative way for event-related brain activity analysis (Sinkkonen *et al* 1995); Tallon-Baudry demonstrated the positive effect of TF analysis in quantifying changes in  $\gamma$ -band oscillatory activity (Tallon-Baudry *et al* 1997). Ryu *et al* used TF analysis on the EEG evoked by negative and positive visual stimuli to distinguish likes from dislikes (Ryu *et al* 1998). All of these successful applications of TF analysis suggest that it may also be a useful tool for ERD and ERS description during motor imagery, which, as we know, are both time and frequency related. Recent studies (Wang and He 2004, Wang *et al* 2004) suggested that ERD/ERS can be well represented by TF distributions, and that TF distribution may provide a suitable means of aiding classification of motor imagery tasks.

In the present study, we have developed a wavelet-based time–frequency scheme for processing and classifying motor imagery signals. The time–frequency property is analyzed by wavelet transformation because of its suitable property for time and frequency localization. ERD/ERS patterns are extracted from the averaged TF distributions, and a classification scheme based on several pairs of symmetric electrodes over the scalp is used to achieve the classification of motor imagery tasks.

## 2. Methods

### 2.1. Experiment paradigm

Nine subjects' EEG signals were recorded during left- or right-hand movement imagination from 59 electrodes, which were placed according to the International 10/20 System. The sampling rate was 100 Hz. The EEG data files consisting of these synchronized imaginary movement experiments were made available by Dr Allen Osman at University of Pennsylvania (Osman and Robert 2001).

180 trial signals were recorded for each subject, 90 on left-hand imagination and 90 on right. Each trial (6 s) began with a blank screen lasting for 2 s. It was designated as a time when it was acceptable for the subject to blink. Then a fixation point replaced this blank screen telling the subject that the trial had begun, which lasted for 0.5 s. After that, at 3.75 s, a time cue (*preparation cue*, lasting 250 ms) of one letter ('L' or 'R') appeared on the screen indicating which hand movement should be imagined; and at 5.0 s another cue 'X' (*execution cue*, lasting 50 ms) appeared, indicating that the subject was to make the requested response. The whole experiment paradigm is shown in figure 1.

### 2.2. Data preprocessing

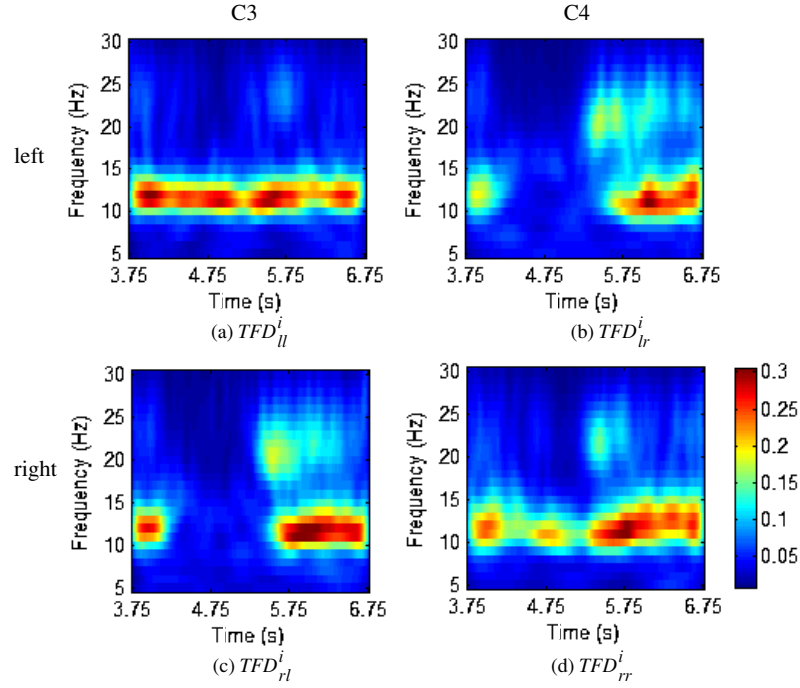
Spatial high pass filters were used in many methods for BCIs as a preprocessing step to accentuate localized activity and reduce diffuse activity. Among them, the surface Laplacian filter, functioned as the second spatial derivative of the instantaneous potential distribution, has been found useful (Hjorth 1975, Perrin *et al* 1987, He and Cohen 1992, Babiloni *et al* 1996, McFarland *et al* 1997b, He 1999, He *et al* 2001, Zhao and He 2001). Assuming that the distances from the four surrounding electrodes to the electrode of interest are approximately equal, the surface Laplacian can be approximately calculated according to the following formula (Hjorth 1975):

$$V_j^{\text{Lap}} = V_j - \frac{1}{4} \sum_{k \in S_j} V_k, \quad (1)$$

where  $V_j$  is the potential of the  $j$ th electrode, and  $S_j$  is an index set of the four surrounding electrodes.

### 2.3. Time–frequency analysis

The waveforms resulting from the Laplacian filter were then subjected to time–frequency (TF) analysis. The goal of TF



**Figure 2.** TFD of channel C3 and C4 for subject 2. Left column is TFD for C3 and right column for C4. The upper row is the TFD of left-hand imagination and the lower one is right-hand imagination. The time period shown here is 3 s from the preparation cue (3.75 s) and frequency band is from 5 to 30 Hz.

analysis was to give a description of the signal energy as a function of time and frequency. A good TF representation aids to extract features correctly and easily.

Based on the uncertainty principle that the more precise the temporal localization, the more inaccurate the frequency information, and vice versa, a compromise between the time and frequency should be considered. In this study, the TF distribution (TFD) was constructed from a Morlet wavelet decomposition which provides an optimal concentration in time and frequency (Tallon-Baudry *et al* 1997, Qin *et al* 2004).

The signal was convoluted by complex Morlet's wavelets  $w(t, f)$ :

$$w(t, f) = A \exp(-t^2/2\sigma_t^2) \exp(2i\pi ft) \quad (2)$$

with  $\sigma_f = 1/2\pi\sigma_t$  and  $A = (\sigma_t\sqrt{\pi})^{-1/2}$ .

A wavelet family was created by setting the trade-off ratio ( $f/\sigma_f$ ) to 7, i.e. the constant that determined the compromise between time and frequency localization. The wavelet family was then convoluted to the signal  $s(t)$  at every frequency  $f$  and the square norm of the convolution was the time-varying energy  $[E(t, f)]$  of the signal at a specific frequency:

$$E(t, f) = |w(t, f) \times s(t)|^2. \quad (3)$$

The TFD was constructed by setting  $f$  from the lowest to the highest frequency of interest. In the present study, a bandwidth from 5 to 30 Hz was chosen and the time window was selected from the preparation cue to 3 s after it, which was 3.75 s to 6.75 s. As shown in figure 2, the horizontal axis refers to time and the vertical refers to frequency. The value of the map represents the power of the signal at a specific time and frequency averaged over all trials of a subject.

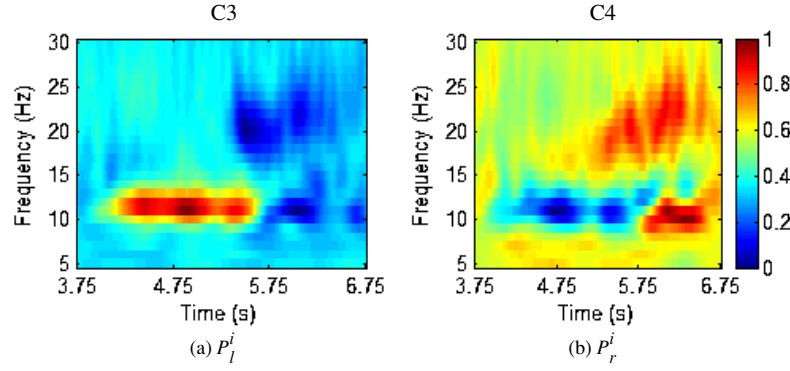
#### 2.4. ERD and ERS

ERD and ERS were used to extract signals representing the motor imagery. Figure 2 displays four averaged TFDs: the first column shows TFDs of electrode C3 and the second column shows those of C4. The first row refers to the TFDs when the subject was asked to imagine left-hand movement and the second row are those corresponding to imagined right-hand movement. In the first row of figure 2, a mu band ERD on electrode C4, the contralateral side to the imagined hand, is clearly observed. A mu band ERD at electrode C3 during the right-hand imaginary movement is also clear in the second row. There is another contralateral beta band ERS—a larger energy on C3 during right-hand movement imagination and on C4 during left-hand movement imagination. There is no obvious beta rhythm ERD, perhaps because this subject did not have this component when performing imaginary movement.

Another important parameter for ERD/ERS patterns—time information—could also be easily extracted from the TFDs. The mu rhythm ERD appeared at the beginning of imagination while beta rhythm ERS occurred at the termination of imagination.

#### 2.5. Weight matrix

The TFDs on the electrodes showed obvious differences during left- and right-hand imaginary movement, and ERD/ERS patterns were easily extracted. In the present study, we used symmetric electrodes (e.g. C3 and C4, C1 and C2, etc) as pairs to extract ERD/ERS patterns. For the  $i$ th electrode pair,



**Figure 3.** ERD and ERS pattern of channel C3 and C4 for subject 2. The mu rhythm shows an ERD pattern, and beta band shows an ERS pattern.

we defined four symbols  $TFD_{ll}^i, TFD_{lr}^i, TFD_{rl}^i, TFD_{rr}^i$ . The first subscript  $l/r$  means the imagined hand and second means the position of the electrode. Taking the set of electrodes C3 and C4 as example, these four symbols correspond to the four TFDs in figures 2(a)–(d).

To find the difference for each electrode during left- and right-hand movement imagination, we subtracted the TFD for each electrode:

$$P_l^i = TFD_{ll}^i - TFD_{rl}^i, \quad P_r^i = TFD_{lr}^i - TFD_{rr}^i. \quad (4)$$

The normalized patterns of  $P_l^i$  and  $P_r^i$  are shown in figure 3. The value corresponding to ERD/ERS was opposite for electrodes on different hemispheres. For example, the mu rhythm ERD had large values for electrode C3 (red) while small values for C4 (blue); while the beta ERS had small values for C3 (blue) while large values for C4 (red).

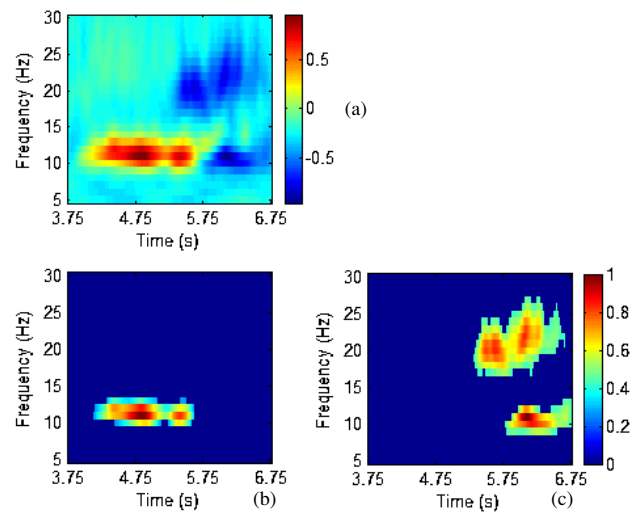
For each pair, we further define a pattern as in the following equation:

$$P^i = P_l^i - P_r^i. \quad (5)$$

This pattern is shown in figure 4(a). As can be seen in this figure, large value corresponds to ERD and small value corresponds to ERS. After setting two thresholds  $\pm 0.25$  and normalizing, we obtained an ERD and an ERS weight matrix, defined as  $WM_{\text{ERD}}^i$  and  $WM_{\text{ERS}}^i$  respectively. The two weight matrices are shown in figures 4(b) and (c), respectively. Figure 4(b) shows the time and frequency information of ERD in the mu band from the beginning of imagination. Figure 4(c) shows the ERS TF information in the beta band at the end of imagination. The alpha band rhythm also shows an ERS after imagination (figure 4(c)).

## 2.6. Classification

The weight matrix can be regarded as a time–frequency window. It was used to time the TFD of a single trial to extract the energy in that window as shown in equation (6). The weighted energy of a trial  $k$  on the  $i$ th pair of electrodes



**Figure 4.** Weight matrix of electrode pair C3 and C4 for subject 2. (a) Weight matrix with both ERD and ERS. (b) Normalized ERD weight matrix. (c) Normalized ERS weight matrix.

was computed according to the equation:

$$\begin{aligned} \bar{E}_{lk}^i &= \sum_{t=3.75}^{6.75} \sum_{f=5}^{30} TFD_{lk}^i(t, f) \times WM_{\text{ERD}}^i(t, f) \\ \bar{E}_{rk}^i &= \sum_{t=3.75}^{6.75} \sum_{f=5}^{30} TFD_{rk}^i(t, f) \times WM_{\text{ERD}}^i(t, f) \\ \bar{E}_{lk}^i &= \sum_{t=3.75}^{6.75} \sum_{f=5}^{30} TFD_{lk}^i(t, f) \times WM_{\text{ERS}}^i(t, f) \\ \bar{E}_{rk}^i &= \sum_{t=3.75}^{6.75} \sum_{f=5}^{30} TFD_{rk}^i(t, f) \times WM_{\text{ERS}}^i(t, f) \end{aligned} \quad (6)$$

where  $WM$  is the normalized weight matrix and the subscript  $l/r$  represents the left/right hemisphere electrode of the  $i$ th electrode pair. So  $\bar{E}_{lk}/\bar{E}_{rk}$  were the weighted energy obtained using ERD weight matrix and  $\bar{E}_{lk}/\bar{E}_{rk}$  were the ERS weighted energy. The difference  $(\bar{E}_{lk}^i - \bar{E}_{rk}^i)$  gave a good description of ERD energy. Because when imagining left-hand movement, a



contralateral ERD made it positive. Similarly, the difference  $(\tilde{E}_{lk}^i - \tilde{E}_{rk}^i)$  gave a good description of ERS energy, as a contralateral ERS made this part negative. And vice versa for right-hand movement imagination. So, it would be easy to just combine the two energy differences and made it as a classification rule.

The energy difference is defined as:

$$ED_k = \sum_{i=1}^{\text{pairs}} (\tilde{E}_{lk}^i - \tilde{E}_{rk}^i) - (\tilde{E}_{lk}^i - \tilde{E}_{rk}^i). \quad (7)$$

It is clear that a left-hand movement imagination will make equation (7) positive while the right-hand imagination will make it negative. So equation (7) would be simple and good for the classification. After getting the energy difference, we only need to know whether it is positive or negative to judge whether that corresponds to left- or right-hand movement imagination.

### 2.7. Cross validation

Tenfold cross validation was performed to test the performance of the proposed procedure. The whole data set of each subject was divided into ten subsets of equal size. The weight matrices were calculated ten times, each time leaving one of the subset out from training and using the omitted subset as a testing group to compute the classification accuracy. The final accuracy was the average of the classification rate obtained each time (i.e. 10 times).

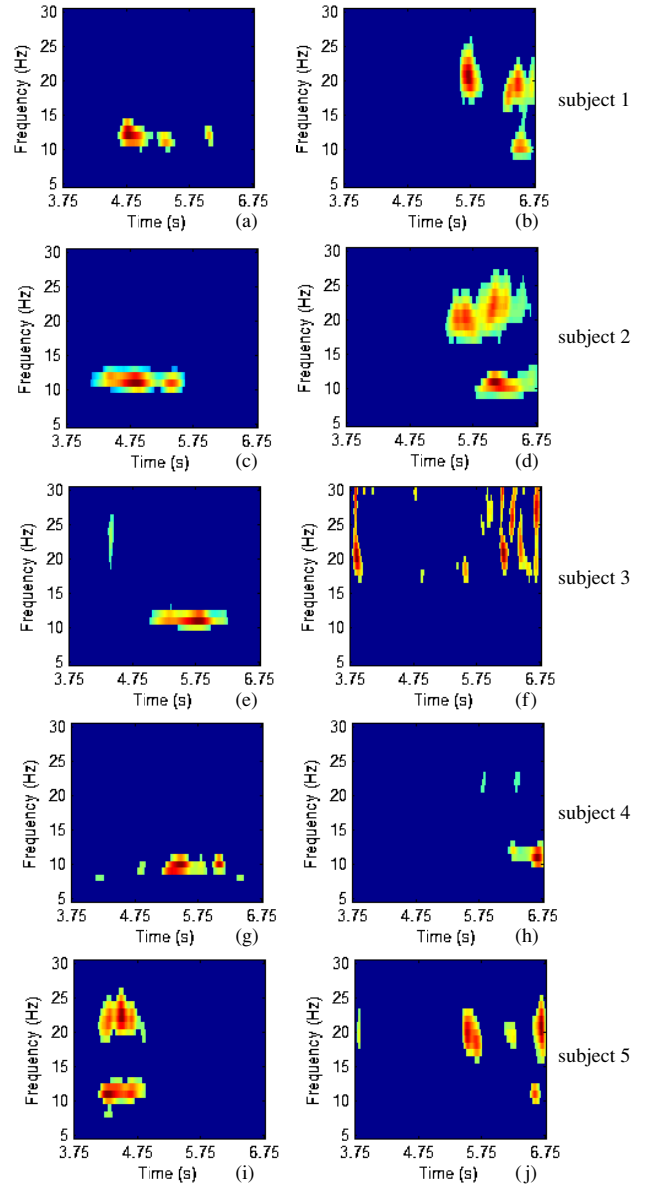
During training, we could get not only the two weight matrices as shown in figures 4(b) and (c), but also the classification accuracy by using just one weight matrix (i.e. ERD only or ERS only). We used only ERD weight matrix (i.e. only the first part of the right side in equation (7)) on the training group and got the classification accuracy  $\tilde{w}^i$ . Then we used just ERS weight matrix (i.e. only the second part of the right side in equation (7)) and got the classification accuracy  $\tilde{w}^i$ . If the subject had a clear ERD/ERS pattern, the classification accuracy must be high based on the corresponding weight matrix. So the accuracy is a good representation of the importance of the pattern and could be used as a weight, as suggested by Wang *et al* (2004). Thus equation (7) could be modified to:

$$ED_k = \sum_{i=1}^{\text{pairs}} \tilde{w}^i (\tilde{E}_{lk}^i - \tilde{E}_{rk}^i) - \tilde{w}^i (\tilde{E}_{lk}^i - \tilde{E}_{rk}^i). \quad (8)$$

Here, the two weights,  $\tilde{w}^i$  and  $\tilde{w}^i$ , can be regarded as a measurement of the importance of ERD and ERS pattern respectively. The classification rule is still the same as that for equation (7), which is that positive equation (8) corresponds to a left-hand movement imagination and negative corresponds to a right-hand movement imagination. So the final accuracy will be defined as the average of the ten testing groups.

## 3. Results

We have illustrated the procedure of getting the ERD and ERS weight matrices for one subject (subject 2) in section 2. The



**Figure 5.** Weight matrices of electrode pair C3 and C4 for all the nine subjects. The left column is the ERD weight matrices and the right column is the ERS weight matrices.

ERD pattern concentrated mainly on the alpha band appearing from the beginning of imagination, and the ERS concentrated on the beta band occurring at the end of imagination. Figure 5 shows the ERD/ERS patterns for all subjects. For subjects 1 and 7, the ERD/ERS patterns were almost the same as subject 2 with an alpha ERD and a beta ERS. Their appearing time was also similar. For subject 5, besides an alpha ERD, a beta ERD was also clear followed by a beta ERS. For subjects 3 and 4, the alpha ERD occurred at time 5 s which was the execution time cue. This may suggest that these two subjects began imagination strictly at this cue. Their ERS patterns were not as concentrated as subjects 1, 2 and 7. The ERS pattern of subject 4 was even harder to see, which was very similar to the ERS pattern of subject 8. Subject 6 showed an alpha

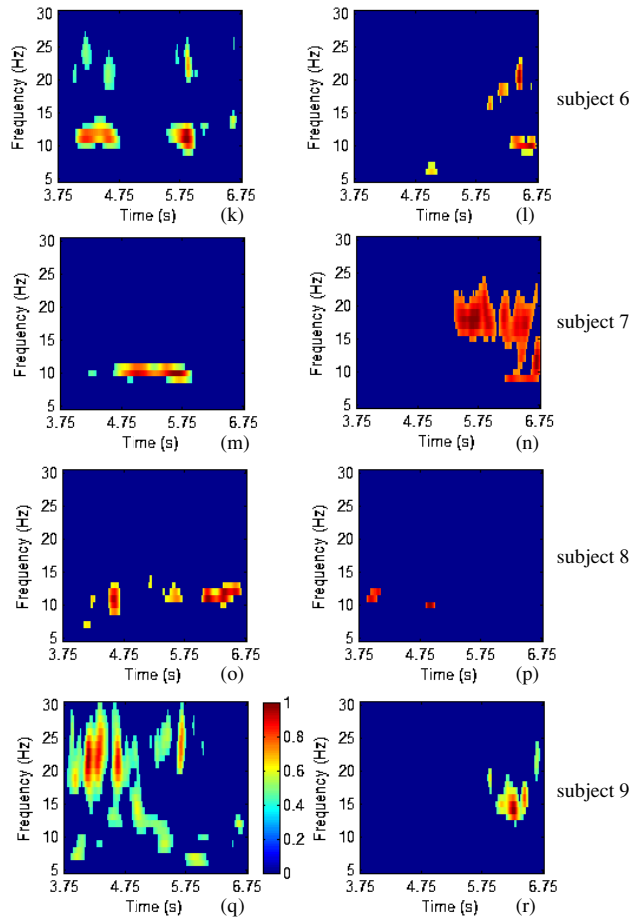


Figure 5. (Continued.)

and beta ERD right after the preparation cue and another ERD after the execution cue, which suggests that this subject may have imagined at both time cues. After imagination, there was an alpha ERS and a beta ERS which could be explained as a rebound after imagination. For subject 9 the beta ERS was concentrated at the end of imagination, while the ERD was scattered along a wide frequency band and a long time period. This diversity of ERD and ERS made it difficult to choose a corresponding frequency band and time window. This was also the limitation of the band-pass filtering method, while in our method we did not need to choose frequency bands. The only thing required was the ERD/ERS pattern which could be easily constructed as introduced in section 2.

We chose several different sets of electrode pairs in the present study. C3 and C4, C1 and C2, overlying the sensorimotor areas, which were activated during left- and right-hand movement imagination, were always used. Several other pairs were added to test their influence on the classification rate. In table 1, the classification rates using different electrode pairs are listed. The first column lists the result with only C3 and C4, C1 and C2; the second column displays the result using C3 and C4, C1 and C2, FC3 and FC4, FC1 and FC2; the third column: C3 and C4, C1 and C2, CP3 and CP4, CP1 and CP2; the fourth column: C3 and C4,

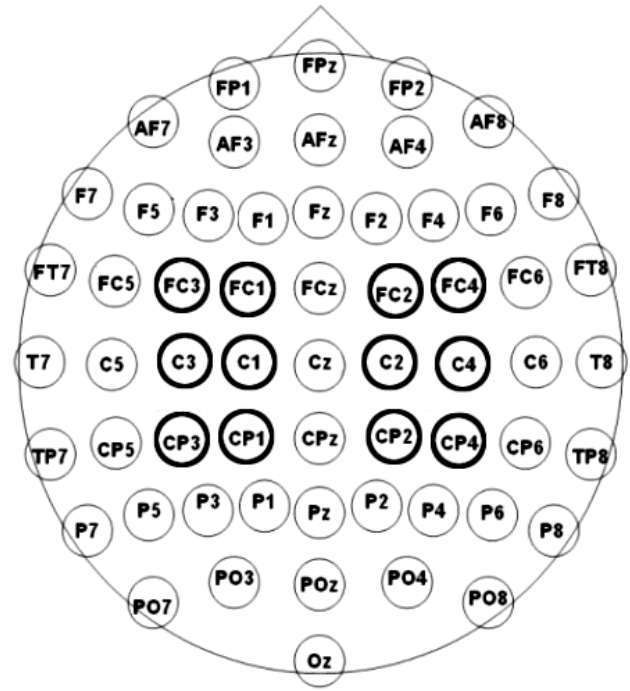


Figure 6. Electrode montage. Bold circles refer to the used pairs.

**Table 1.** Classification accuracy using different electrode pairs. Pairs I: C3 and C4, C1 and C2. Pairs II: C3 and C4, C1 and C2, FC3 and FC4, FC1 and FC2. Pairs III: C3 and C4, C1 and C2, CP3 and CP4, CP1 and CP2. Pairs IV: C3 and C4, C1 and C2, FC3 and FC4, FC1 and FC2, CP3 and CP4, CP1 and CP2. Pairs V: C3 and C4, C1 and C2, FC1 and FC2, CP1 and CP2.

|           | Pairs I | Pairs II | Pairs III | Pairs IV | Pairs V |
|-----------|---------|----------|-----------|----------|---------|
| Subject 1 | 72.8    | 62.8     | 67.2      | 62.8     | 68.3    |
| Subject 2 | 91.7    | 88.9     | 90.0      | 90.1     | 90.0    |
| Subject 3 | 69.4    | 68.3     | 73.9      | 74.4     | 73.9    |
| Subject 4 | 68.3    | 65.6     | 69.4      | 71.7     | 73.3    |
| Subject 5 | 73.9    | 75.0     | 68.3      | 72.2     | 76.7    |
| Subject 6 | 72.8    | 76.1     | 75.6      | 78.3     | 76.1    |
| Subject 7 | 81.7    | 89.4     | 75.0      | 89.4     | 90.6    |
| Subject 8 | 56.7    | 68.3     | 61.7      | 64.4     | 68.3    |
| Subject 9 | 85.7    | 82.8     | 85.0      | 84.4     | 86.1    |
| Average   | 74.8    | 75.3     | 74.0      | 76.5     | 78.2    |

C1 and C2, FC3 and FC4, FC1 and FC2, CP3 and CP4, CP1 and CP2, and the fifth column: C3 and C4, C1 and C2, FC1 and FC2, CP1 and CP2 (refer to figure 6 for the montage). The fourth composition had the most pairs but did not return the best result. The fifth composition gave the best average classification accuracy for the nine subjects, which suggests that the more electrode pairs may not lead to better results. Some pairs did not positively contribute to the whole energy difference, only noise or even negative influence.

From table 1, it is obvious that the electrode position plays an important role for the final classification. For subject 1, if only two pairs, C3 and C4, C1 and C2, were used, the accuracy was 72.8%; if two other pairs, FC3 and FC4, FC1 and FC2, were added in, the accuracy dropped to 62.8%, a 10% decrease. While for subject 8, the same two pairs, FC3 and FC4, FC1

and FC2, would result in a 12% increase in the classification accuracy from 56.7% to 68.3%. The similar phenomenon occurred when using pairs CP3 and CP4, CP1 and CP2. They resulted in a higher accuracy for subject 3 and a lower accuracy for subject 7 as compared to the results using only pairs C3 and C4, C1 and C2. When all six pairs were used, the results were not always the best among the five organizations. Only for subjects 3 and 6 the six pairs functioned best. The remaining seven subjects failed to demonstrate better performance. This diversity in the performance of pairs on different subjects suggests the varied topography of ERD/ERS pattern among subjects, which was also a difficulty in ERD/ERS-based motor imagery classification.

#### 4. Discussion

In the present study, we have demonstrated that time–frequency analysis is an efficient way for motor imagery EEG signal preprocessing and classification. The TFD constructed by wavelet decomposition contains both the time and frequency information and thus provides a more preferred description of the signal than spectral analysis and band-pass filtering methods.

The spectral analysis method used by Wolpaw and colleagues (Wolpaw *et al* 1991, 1997, 2002) uses an AR model and calculates the band power in several frequency bands to define the control signals (McFarland *et al* 1997a, 1997b, 2000). Their method didn't use the time domain information, and the manually chosen frequency band may lose some other frequency bands' information. Babiloni *et al* used the Welch periodogram to estimate the power spectral density (PSD) on the surface Laplacian transformed data and then used signal space projection (SSP) method for classification (Babiloni *et al* 2000), which also didn't utilize the time information. Pfurtscheller and colleagues used DSLVQ to select the frequency bands for each subject (Kohonen 1990). For the selected bands, the band power in several time segments was calculated and classified by LVQ (Kalcher *et al* 1996). The DSLVQ method, while it may choose the most reactive frequency, might still lose some information if the reactive frequency band is wide (like subject 9 in this study). For band power calculation on each time segment, all the sample points were treated the same which was not fair for different sample points. Wang and He used band-pass filter to decompose the signal into many narrow-banded sub-signals and used PCA on the down-sampled sub-signals (Wang and He 2004). They then set different weights for LDA to classify left- and right-hand imaginary movement signals (Wang and He 2004). This down sampling and PCA may also lose some time domain information. Another method developed by Wang and He also band-pass filter the signal into several bands. Then in each band, they get the average energy of 14 even time intervals. By weighting these frequency and time segments according to their importance, they got good performance. However, this method also didn't catch enough detail in time domain by averaging the powers. In the present method, by using time–frequency analysis, we can get both time and frequency information simultaneously; thus all the

useful information were included in the time–frequency map. We also do not need to pick up any frequency bands or time segments which might cause loss of information. The entire alpha and beta frequency range and 3 s time signal were used for analysis.

Of interest is another preliminary idea in this method about the proper use of electrode pairs. The energy located in the ERD/ERS window of electrode pairs has been compared. Based on the fact that mental imagery of hand movement produces an energy decrease (ERD) and an energy increase (ERS) over the cortical areas contralateral to the imaginary moved hand, we could judge position of the imagined hand. If the energy difference between two hemispheres is positive, the imaginary hand would be left; otherwise, it would be right. With the help of the two patterns (i.e. weight matrices), the energy difference provides a so clear and simple pattern that this very simple decision rule can function effectively in the present analysis. The average classification rate over the nine subjects reached 78.2%, and two subjects reached a 90% classification rate. Comparing with previously reported analysis of the same data set (76–77%) (Sajda *et al* 2003, Wang and He 2004), the present results are promising, suggesting the usefulness of the present TF method for aiding signal processing and classification of motor imagery tasks.

In the present study, we have used the electrode pairs with symmetric electrodes, such as C3 and C4, C1 and C2. This is based on an assumption that the response of the left- or right-hand movement imagination resulted in an opposite energy topography change on the brain. In other words, if the left-hand movement imagination produces an ERD on C4, a right-hand movement imagination is expected to produce the same ERD on C3. Even though this coincides with our common sense and the connection in the brain is symmetric, a question remains that if ERD/ERS patterns are also symmetric. This is not so clear. So to ameliorate the method of choosing the electrode pairs is expected to further improve the classification accuracy.

Another good feature of the present method is its simplicity. There is no need for complex classifiers or a manually chosen frequency band. Additionally, the computation of TFD is very easy. These all make it a promising method for an online EEG-based brain–computer interface.

In summary, we have developed a method to conveniently and accurately extract the ERD/ERS patterns during motor imagery, and tested its applicability to classification of motor imagination in nine human subjects. This wavelet-based method, by combining both time and frequency information, gives a full view of ERD/ERS pattern and helps ease the classification step. The present results are promising and suggest time–frequency analysis merits further investigation for brain computer interface research.

#### Acknowledgments

The authors are grateful to Dr Allen Osman of University of Pennsylvania for making his data available, Zhongming Liu for useful discussions and Christopher Wilke for proofreading



the manuscript. This work was supported in part by NSF BES-0218736, NSF BES-0411898 and NIH RO1EB00178.

## References

- Babiloni F, Babiloni C, Carducci F, Fattorini L, Onorati P and Urbano A 1996 Spline Laplacian estimate of EEG potentials over a realistic magnetic resonance-constructed scalp surface model *Electroencephalogr. Clin. Neurophysiol.* **98** 363–73
- Babiloni F, Cincotti F, Bianchi L, Pirri G, Millán JdR, Mouriño J, Salinari S and Marciani M G 2001 Recognition of imagined hand movements with low resolution surface Laplacian and linear classifiers *Med. Eng. Phys.* **23** 323–8
- Babiloni F, Cincotti F, Lazzarini L, Millán J, Mourino J, Varsta M, Heikkinen J, Bianchi L and Marciani M G 2000 Linear classification of low-resolution EEG patterns produced by imagined hand movements *IEEE Trans. Rehabil. Eng.* **8** 186–8
- Birbaumer N and Kubler A 2000 The thought translation device (TTD) for completely paralyzed patients *IEEE Trans. Rehabil. Eng.* **8** 190–3
- Farwell L A and Donchin E 1988 Talking off the top of your head: toward a mental prosthesis utilizing event-related brain potentials *Electroencephalogr. Clin. Neurophysiol.* **70** 510–23
- He B 1999 Brain electric source imaging: scalp Laplacian mapping and cortical imaging *Crit. Rev. Biomed. Eng.* **27** 149–88
- He B and Cohen R J 1992 Body surface Laplacian ECG mapping *IEEE Trans. Biomed. Eng.* **39** 1179–91
- He B, Lian J and Li G 2001 High-resolution EEG: a new realistic geometry spline Laplacian estimation technique *Clin. Neurophysiol.* **112** 845–52
- Hjorth B 1975 An on-line transformation of EEG scalp potentials into orthogonal source derivations *Electroencephalogr. Clin. Neurophysiol.* **39** 526–30
- Kalcher J, Flotzinger D, Neuper C, Golly S and Pfurtscheller G 1996 Graz brain–computer interface III—toward communication between humans and computers based on online classification of three different EEG patterns *Med. Biol. Eng. Comput.* **34** 382–8
- Kohonen T 1990 The self-organizing map *Proc. IEEE* **78** 1464–80
- Kordylewski H, Graupe D and Kai L 2001 A novel large-memory neural network as an aid in medical diagnosis applications *IEEE Trans. Inf. Technol. Biomed.* **5** 202–9
- McFarland A D J, Lefkowitz A T and Wolpaw J R 1997a Design and operation of an EEG-based brain–computer interface with digital signal processing technology *Behav. Res. Methods Instrum. Comput.* **29** 337–45
- McFarland D J, McCane L M, David S V and Wolpaw J R 1997b Spatial filter selection for EEG-based communication *Electroencephalogr. Clin. Neurophysiol.* **103** 386–94
- McFarland D J, Miner L A, Vaughan T M and Wolpaw J R 2000 Mu and beta rhythm topographies during motor imagery and actual movements *Brain topogr.* **12** 177–86
- Millán JdR, Mouriño J, Franzé M, Cincotti F, Varsta M, Heikkinen J and Babiloni F 2002 A local neural classifier for the recognition of EEG patterns associated to mental tasks *IEEE Trans. Neural Netw.* **13** 678–86
- Muller-Gerking J, Pfurtscheller G and Flyvbjerg H 1999 Designing optimal spatial filters for single-trial EEG classification in a movement task *Clin. Neurophysiol.* **110** 787–98
- Nigam V P and Graupe D 2004 A neural-network-based detection of epilepsy *Neurol. Res.* **26** 55–60
- Osman A and Robert A 2001 Time-course of cortical activation during overt and imagined movements *Cognit Neuroscience Annual Meeting (New York, March 2001)*
- Perrin F, Bertrand O and Pernier J 1987 Scalp current density mapping: value and estimation from potential data *IEEE Trans. Biomed. Eng.* **34** 283–8
- Pfurtscheller G 1999 Event-related Desynchronization *Handbook of Electroencephalography and Clinical Neurophysiology* vol 6 (The Netherlands: Elsevier) pp 51–65
- Pfurtscheller G and Neuper C 1994 Event-related synchronization of mu rhythm in the EEG over the cortical hand area in man *Neurosci. Lett.* **174** 93–6
- Pfurtscheller G, Neuper C, Flotzinger D and Pregenzer M 1997 EEG-based discrimination between imagination of right and left hand movement *Electroencephalogr. Clin. Neurophysiol.* **103** 642–51
- Qin L, Ding L and He B 2004 Motor imagery classification by means of source analysis for brain computer interface applications *J. Neural Eng.* **1** 135–41
- Ryu C S, Song Y, Kim S H, Yi I, Kim J and Sohn J 1998 A time–frequency analysis of the EEG evoked by negative and positive visual stimuli *Proc. 20th IEEE Eng. Med. Biol. Soc.* **20** pp 105–8
- Sajda P, Gerson A, Mueller K R, Blankertz B and Parra L 2003 A data analysis competition to evaluate machine learning algorithms for use in brain–computer interfaces *IEEE Trans. Neural. Syst. Rehabil. Eng.* **11** 184–5
- Sinkkonen J, Tiitinen H and Näätänen R 1995 Gabor filters: an informative way for analyzing event-related brain activity *J. Neurosci. Methods* **56** 99–104
- Sutter E E 1984 The visual evoked response as a communication channel *Proc. IEEE/NSF Symp. Biosensors* pp 95–100
- Tallon-Baudry C, Bertrand O, Delpuech C and Pernier J 1997 Oscillatory  $\gamma$ -band activity induced by a visual search task in humans *J. Neurosci.* **17** 722–34
- Vallabhaneni A and He B 2004 Motor imagery task classification for brain computer interface applications using spatiotemporal principle component analysis *Neurol. Res.* **26** 282–7
- Vallabhaneni A, Wang T and He B 2005 Brain computer interface *Neural Engineering* ed B He (New York: Kluwer/Plenum) pp 85–122
- Wang T, Deng J and He B 2004 Classifying EEG-based motor imagery tasks by means of time–frequency synthesized spatial patterns *Clin. Neurophysiol.* **115** 2744–53
- Wang T and He B 2004 An efficient rhythmic component expression and weighting synthesis strategy for classifying motor imagery EEG in brain computer interface *J. Neural Eng.* **1** 1–7
- Wolpaw J R, Birbaumer N, McFarland D J, Pfurtscheller G and Vaughan T M 2002 Brain–computer interfaces for communication and control *Clin. Neurophysiol.* **113** 767–91
- Wolpaw J R, Flotzinger D, Pfurtscheller G and McFarland D J 1997 Timing of EEG-based cursor control *J. Clin. Neurophysiol.* **14** 529–38
- Wolpaw J R, McFarland D J, Neat G W and Forneris C A 1991 An EEG-based brain–computer interface for cursor control *Electroencephalogr. Clin. Neurophysiol.* **78** 252–9
- Zhao F and He B 2001 A new algorithm to estimate surface Laplacian and its applications to visual evoked potentials *Electromagnetics* **21** 633–40